


Cartesian Product $X \times Y$, $\{(x, y) \mid x \in X, y \in Y\}$

A relation R on $X \times Y$, $R \subseteq X \times Y$

A binary relation $R \subseteq X \times X$

Examples: $X = \mathbb{R}$, one binary relation on $\mathbb{R}$, $\geq$

- $X = \mathbb{R}$, $>$

- $X = \{\text{students in this class}\}$, $R = \text{"taller than"}$

$(\mathbb{R}, \geq)$

$\geq$, transitive, complete

For the greater than relation $\geq$, the derived symmetric part of $\geq$ is "=", $a, b \in \mathbb{R}, a \geq b, b \geq a \Leftrightarrow a = b$

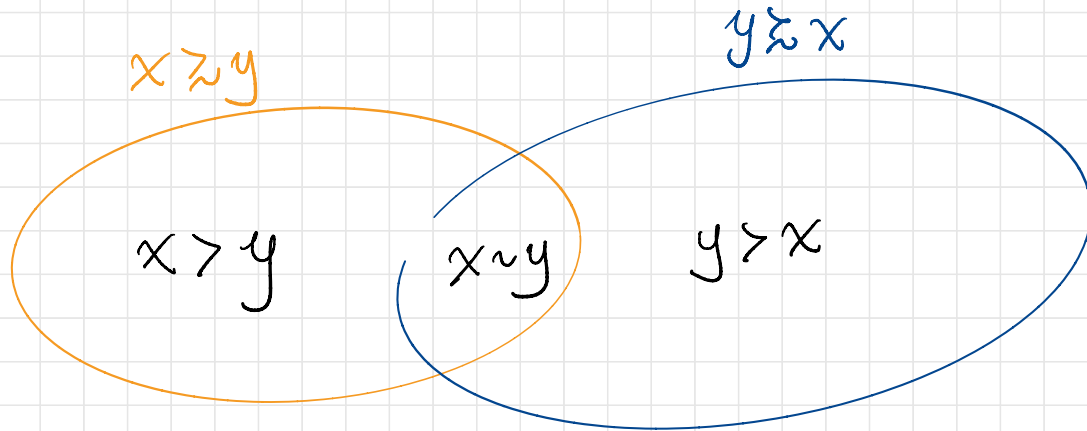
The asymmetric part of $\geq$ is the relation $>$;

why? If $a, b \in \mathbb{R}, a \geq b, \neg(b \geq a) \Leftrightarrow a > b$

Given $\succeq$ a preference relation

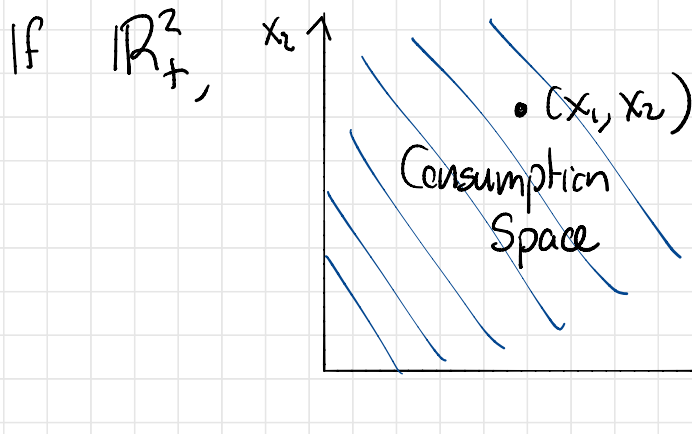
Symmetric part $x \sim y \Leftrightarrow (x \succeq y \text{ \& } y \succeq x)$

Asymmetric part $x > y \Leftrightarrow (x \succeq y \text{ \& } \neg y \succeq x)$



Lem1. If X is finite, we have at least two elements $\bar{x}, \underline{x}$ such that, for every $y \in X$, $\bar{x} \succeq y \succeq \underline{x}$

X infinite? Most of Consumer Choice/Behavior is carried out in $\mathbb{R}_+^n$, $(x_1, x_2, \dots, x_n) \in \mathbb{R}_+^n$
consumption bundle



Lexicographic Preference in $[0,1] \times [0,1]$

$$x \succ y \Leftrightarrow (i) \underset{\text{or}}{x_1 > y_1}$$

$$(ii) x_1 = y_1, x_2 > y_2$$

Note: $\underset{\uparrow}{\succ}$ is different from $\underset{\uparrow}{\succ}$

Lexicographic
P.R.

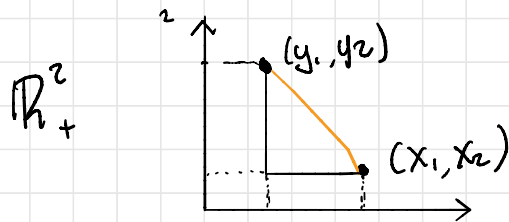
strictly greater
than

Symmetric part of Lexicographic Preference Relation:

$$x \sim y \Leftrightarrow \left(\begin{array}{l} x_1 = y_1 \text{ and} \\ x_2 = y_2 \end{array} \right) \Leftrightarrow x = y$$

Take $X = \mathbb{R}_+^n$, we need a notion of distance between two elements $x, y \in \mathbb{R}_+^n$

- Euclidean distance : $d(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2}$

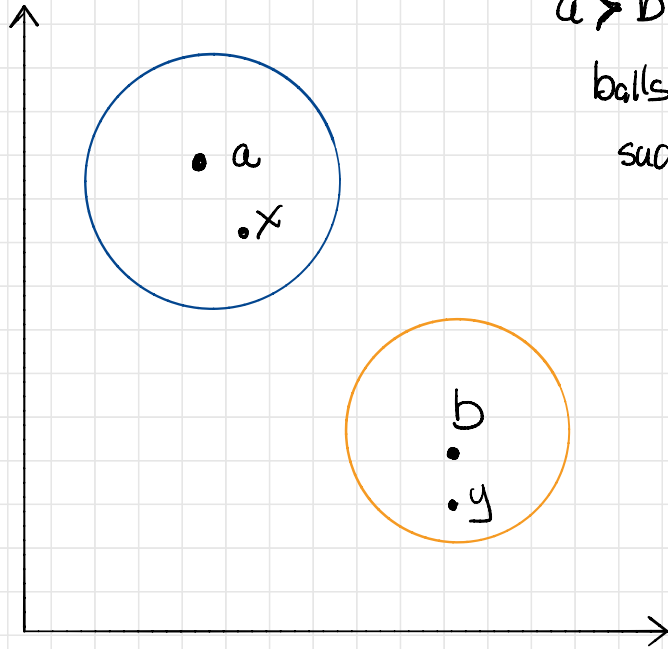


- Taxicab / Manhattan distance : $d(x, y) = \sum_{i=1}^n |x_i - y_i|$

With a notion of distance, we can define the ball around $a \in X$ of radius $r > 0$ as $B(a, r)$

$$B(a, r) = \{y \in \mathbb{R}^n : d(a, y) < r\}$$

Continuity: (the $\mathbb{R}^2_+$ graph is for)



$a \succ b$ if for any $r > 0$, there are
balls $B(a, r)$ and $B(b, r)$
such that for all
 $x \in B(a, r)$
 $y \in B(b, r)$ $\Rightarrow x \succ y$